Files & Paths Cheat Sheet

Mac & Windows

Find a file, read it, write the result. Python for Biologists, Lab 4.

Run your script — the same way every time

1. VS Code → **File** → **Open Folder** → your project folder (e.g. `lab4`).
2. Terminal inside VS Code: `Ctrl` `` ` `` — check it with `pwd`: it should end in your folder.
3. `uv run peek.py` — the script runs with the terminal's folder as its **current working directory**.

Paths are objects

```
from pathlib import Path

# the folder this script is saved in
here = Path(__file__).parent
# / joins path parts
data_dir = here / "sequences"
bee = data_dir / "Bombus_terrestris.fasta"

bee.name      # 'Bombus_terrestris.fasta'
bee.stem      # 'Bombus_terrestris'
bee.suffix    # '.fasta'
bee.parent    # the sequences folder
bee.exists()  # True / False
bee.resolve() # full absolute path
```

Terminal → Python

TERMINAL	PYTHON
`pwd`	`Path.cwd()`
`ls`	`list(p.iterdir())`
`ls *.fasta`	`sorted(p.glob("*.fasta"))`
`cd sequences`	`p / "sequences"`
`cd ..`	`p.parent`
`mkdir results`	`p.mkdir(exist_ok=True)`

Reading

```
# the file closes when the block ends
with open(bee) as f:
    text = f.read()        # one string

with open(bee) as f:
    lines = f.read().splitlines()
    # a list of lines, no \n

with open(bee) as f:
    for line in f:     # line by line
        line = line.strip()  # drop \n

text = bee.read_text()   # shortcut
```

Everything you read is a **string** — `float("24.0")` before maths.

Loop over a folder

```
files = sorted(data_dir.glob("*.fasta"))
for fasta_file in files:
    species = fasta_file.stem
    dna = read_fasta(fasta_file)
    print(species, len(dna))
```

Writing

```
out = here / "results"
out.mkdir(exist_ok=True)

with open(out / "bees.csv", "w") as f:
    f.write("species,gc\n")  # header
    for species, gc in rows:
        f.write(f"{species},{gc}\n")
```

MODE	DOES
`"r"`	read (default)
`"w"`	empties the file first!
`"a"`	add to the end

`write()` adds no newline and takes only strings.

When it goes wrong

- **FileNotFoundError** — print `path.resolve()`. Typo, or a relative path from the wrong folder? Use `Path(__file__).parent`.
- **IsADirectoryError** / **PermissionError** — you opened a folder, or (Windows) Excel has the file open.
- **Everything on one line?** Missing `\n`.
- **Results file empty?** `"w"` inside a loop wipes it each time — open once, before the loop.
- **Windows path in a string?** `"C:/Users/…"` or `r"C:\Users\…"`.